

Project Initialization and Planning Phase

Date	15 July 2026
Team ID	SWTID-2026-1518
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to develop an intelligent credit card approval prediction system using machine learning to improve the speed, accuracy, and consistency of credit approval decisions. The system analyzes applicant information to predict approval eligibility, reducing manual effort, minimizing risk, and supporting better decision-making.

Project Overview	
Objective	The primary objective is to develop a machine learning model that predicts whether a credit card application should be approved based on applicant data, enabling faster and more reliable decisions.
Scope	The project focuses on analyzing applicant information, training predictive models, and providing an efficient credit card approval prediction system for financial institutions.
Problem Statement	
Description	Traditional credit card approval processes are often time-consuming and prone to inconsistent decision-making. An automated prediction system is required to improve efficiency and accuracy.
Impact	The solution helps reduce processing time, improves prediction accuracy, minimizes financial risk, and enhances customer experience.
Proposed Solution	
Approach	Using machine learning algorithms to analyze applicant attributes and predict credit card approval with high accuracy.
Key Features	- Machine learning-based credit card approval prediction model.

	Real-time prediction for instant approval decisions. Accurate risk assessment using applicant data.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv